

CORE SETUP, UNITS, WATER BALANCE [M2, M7]

System logic: define boundary + time interval; classify each term as input, output, or storage. Conservation: $\Delta S = I - O$. Positive ΔS = gain; negative ΔS = loss.

Full balance: $\Delta S = (P + Q_{in} + G_{in}) - (ET + Q_{out} + G_{out})$

Steady state: $\Delta S = 0$, therefore Σ inputs = Σ outputs. Steady state does *not* mean no flow.

Depth to volume: $V = dA$. **Rate to volume;** $V = Qt$. **Level change:** $\Delta h = \Delta S/A$.

Unit ladder: 1 m = 100 cm = 1000 mm; 1 km² = 10⁶ m²; 1 m³ = 1000 L; 1 day = 86,400 s; 1 L water $\approx$ 1 kg.

Fast conversion: 1 mm over 1 km² = 1000 m³. Never add mm/day to m³/day. Convert depth to m, multiply by area, then combine volume rates.

Depth-volume shortcuts: 1 mm over 2, 4, or 5 km² gives 2,000, 4,000, or 5,000 m³. For a net depth, subtract depths first when they apply to the same area.

Quiz pattern: 5 km² $\times$ (12 - 8) mm/day = 20,000 m³/day. Net surface + groundwater = (23 + 5) - (16 + 3) = +9, so ΔS = +20,009 m³/day.

Lake rise pattern: 31,000 m³/day $\div$ 2,000,000 m² = 0.0155 m/day = 15.5 $\approx$ 16 mm/day.

Useful rearrangements

At steady state: $P + Q_{in} + G_{in} = ET + Q_{out} + G_{out}$

Unknown ET: $ET = P + Q_{in} + G_{in} - Q_{out} - G_{out} = \Delta S$.

Unknown Q_{in} : $Q_{in} = \Delta S + ET + Q_{out} + G_{out} - P - G_{in}$.

Calculation routine

1) Write equation. 2) Mark signs/arrows. 3) make SI units and time bases compatible. 4) substitute with units. 5) interpret sign and magnitude. 6) round last.

ATMOSPHERIC MOISTURE + SNOW [M3]

Relative humidity: RH = (e/e_s) $\times$ 100%. Saturated when e = e_s, RH = 100%.

Vapour pressure deficit: VPD = e_s - e. VPD > 0: evaporation favoured; = 0: equilibrium; < 0: condensation favoured.

Cooling lowers temperature-dependent e_s. If actual e stays fixed, RH rises until the dew point where e = e_s.

RH check: e = 2.0 kPa, e_s = 5.0 kPa $\rightarrow$ RH = 0.40 = **40%**. At 18°C, course graph e_s $\approx$ 2.0 kPa; e = 1.0 kPa gives 50% RH and dew point $\approx$ 7°C.

Adiabatic change: $\Delta T = -\Gamma \Delta z$. Course rates: dry 1.0°C/100 m; wet 0.6°C/100 m; environmental 0.65°C/100 m.

Lapse check: dry air rising 1200 m cools 12°C; saturated air rising 1200 m cools 7.2°C. Descending air warms by the applicable rate.

Snow: density ratio D_s = SWE/snow depth; SLR = snow depth/SWE = 1/D_s; SWE = depth/SLR.

Fresh snow density ratio $\approx$ 0.06 (6%); wind-packed snow may reach $\geq$ 0.30. Wind redistributes snowfall, so ground snow depth can be highly variable.

Snow check: 20 cm snow at 10:1 SLR contains 2 cm SWE. A 20:1 pack has less water per unit snow depth than a 2:1 pack.

PRECIPITATION DATA FORMULAS [M4]

Intensity: I = P/ Δt , commonly mm/h. **Snow core:** V = Ad; $\rho = m/V$.

Missing station: P = (N/n) Σ (P/N_i). N = normal annual precipitation.

Arithmetic mean: $\bar{P} = (1/G) \Sigma p_g$. Best for flat terrain, even gauges, and spatially uniform precipitation.

Thiessen: $\bar{P} = \Sigma w_i P_i$, where $\Sigma w_i = 1$. **Isohyetal:** $V = \Sigma(A_{band} \bar{P}_{band})$.

Double mass: K = slope after change / slope before change. Use the course-specified correction direction.

Method ranking: arithmetic = equal weight; Thiessen = station-area weight; isohyetal = contours/bands and usually best where spatial variation or topography matters. Always report depth units after dividing total volume by total area.

VARIABLE KEY: BALANCE, AIR, PRECIPITATION

Balance: ΔS = storage change; I/O = total inputs/outputs; P = precipitation; ET = evapotranspiration; $Q_{in/out}$ = surface-water inflow/outflow; $G_{in/out}$ = groundwater inflow/outflow.

Geometry/time: V = volume; d or Δh = depth or level change; A = area; Q = discharge rate; t or Δt = elapsed time. Context distinguishes I = total input from I = rainfall intensity.

Air: RH = relative humidity; e or e_s = actual vapour pressure; e_s = saturation vapour pressure; VPD = vapour pressure deficit; Γ = lapse rate; Δz = elevation change.

Snow: D = snow density ratio; SWE = snow water equivalent; SLR = snow-liquid ratio; ρ = density; m = mass.

Gauge/area: P_i = missing-station estimate; N_s, N_n = normal annual P at missing/nearby stations; P_n = observed nearby P; n = number of nearby stations; G = gauge count; p_g = gauge P; w_i = Thiessen area fraction; A_{band} and $\bar{P}_{band}$ = isohyetal-band area and mean depth; K = double-mass factor.

EVAPORATION + TRANSPIRATION [M5]

Evaporation: liquid to vapour. **Sublimation:** solid to vapour. **Transpiration:** plant vapour loss through stomata. **ET:** evaporation from free water, interception, soil, etc. + transpiration.

Energy is usually the strongest control. Warm, dry, sunny, windy conditions favour E/ET. Wind removes humid boundary air but cannot exceed the radiation/energy-limited maximum.

Fresh water evaporates faster than saline water. Shallow water warms quickly and has high summer rates; deep water stores heat and may maintain evaporation at night/cool seasons. Moving air humidifies across a long lake, reducing average rate downwind even when total volume loss is larger.

Soil evaporation rises with moisture, a shallow water table, and dark colour. Vegetation reduces soil evaporation by shade, wind blockage, and humid near-surface air.

Stomata opening duration depends mainly on solar energy/day length, temperature, and RH. Hydrophyte: roots in water; mesophyte: soil moisture; xerophyte: arid, shallow dense roots; phreatophyte: roots reach water table.

MEASURING ET [M6]

Potential ET: maximum with unlimited water and unsaturated air. **Actual ET:** current conditions; actual $\leq$ potential.

Pan: E_{pan} = depth loss/time. Adjusted E = (depth loss $\times$ K_p)/time. Correct for rain and added/removed water.

Lysimeter: d = V/A after converting mass loss to water volume. Weighing type uses mass loss; water-balance type tracks levels and flows.

Energy balance: R - R_E - R_A = H_E + H_A + H_B + H_C. Simplified course form: H_E = R - R_E - R_A.

ET water balance: E = P + SW_i + GW_i - SW_o - GW_o - ΔS .

Empirical: E = K f(u)(e_s - e_a)ⁱⁿ_v. Local calibration limits transfer to unlike sites.

Groundwater fluctuation: $\bar{E} = S_y(24a + b)$. S_y = effective specific yield; a = night recovery rate; b = signed 24 h net water-table change.

Worked GW ET: a = (0.60 - 0.50 m)/4 h = 0.025 m/h; b = 0.50 - 0.62 = -0.12 m; S_y = 0.13. ET = 0.13[24(0.025) - 0.12] = 0.0624 m/day = **62.4 mm/day**.

Method choice: pan = potential open-water E; lysimeter = actual sample ET; Penman-Monteith = potential ET model; chambers capture local vapour; groundwater method requires a shallow plant-accessible water table.

Pan correction pattern: apply K_p to the corrected depth loss, then divide by elapsed time. **Chamber pattern:** vegetated chamber loss - comparable covered bare-surface loss = transpiration.

Energy terms: incoming radiation is divided among reflected radiation, emitted longwave radiation, latent heat of evaporation, sensible air/surface heating, and plant growth.

SOIL PHASE RELATIONSHIPS [M8]

Volumes: V = V_t + V_v; V_v = V_s + V_w. **Mass:** m_t = m_s + m_w.

Porosity: n = V_v/V_t. **Void ratio:** e = V_v/V_s. Check denominator.

Gravimetric water content: w = m_w/m_s. **Saturation:** S = V_w/V_v.

Bulk density: $\rho = m/V$. **Plasticity index:** PI = LL - PL.

Worked checks: V_t = 18 cm³, V_v = 60 cm³ $\rightarrow$ n = 0.30 = 30%. V_v = 15, V_s = 50 $\rightarrow$ e = 0.30. LL = 54%, PL = 22% $\rightarrow$ PI = 32%.

Water content example: m_s = 8 g and m_w = 40 g $\rightarrow$ w = 0.20 = 20%. Saturation example: V_w = 5 cm³, V_v $\geq$ 10 cm³ $\rightarrow$ S_r = 50%.

With increasing water content: solid $\rightarrow$ semi-solid $\rightarrow$ plastic $\rightarrow$ liquid; boundaries are shrinkage limit, plastic limit, liquid limit. Low PI = small moisture change can trigger liquid behaviour.

Critical distinction: porosity = amount of void space; permeability = ease of flow through connected pores. Clay can have high porosity but low permeability and high retention.

VARIABLE KEY: ET + SOIL

ET methods: E or ET = evaporation/evapotranspiration rate; E_{pan} = pan rate; K_p = pan coefficient; SW/GW = surface-water/groundwater flux; ΔS = storage change.

Energy/model: R = incoming radiation; R_E = reflected shortwave; R_s = outgoing longwave; H_E = latent heat used by evaporation; H_A, H_B, H_C = air warming, surface warming, plant growth; K = empirical coefficient; f(u) = wind function; u = wind speed.

GW fluctuation: S_y = effective specific yield; a = midnight-to-4 a.m. water-table recovery rate; b = signed water-level change over exactly 24 h; 24 converts hourly a to one day.

Soil volumes: V_t = total; V_s = solids; V_v = voids; V_w = water; V_a = air. **Masses:** m_t = total; m_s = dry solids; m_w = water.

Soil indices: n = porosity; e = void ratio; w = gravimetric water content; S = degree of saturation; ρ = bulk density; LL/PL = liquid/plastic limits; PI = plasticity index.

GROUNDWATER CALCULATIONS [M9]

Elevations: z_{wt} = z_{ground} - d_{water}; z_b = z_{ground} - d_{well}. Use one vertical datum. **Hydraulic head:** h = z + h_p; pressure head h_p = h - z_b. **Hydrostatic pressure:** p = ρgh_p (Pa; $\approx$ 1000 for kPa).

Pressure check: 10 m water column gives p = (1000 kg/m³)(9.8 m/s²)(10 m) = 98,000 Pa = **98 kPa**.

Hydraulic gradient: i = $\Delta h/L$. Flow is from high to low head; in homogeneous material, flow lines cross equipotentials at 90°.

Darcy specific discharge: q = Ki. **Total discharge:** Q = qA = KiA.

Pore-water velocity: v = q/n = Ki/n. **Travel time:** t = L/|v|. Convert porosity % to decimal.

Course check: K = 2.16 m/day, i = 0.030 $\rightarrow$ q = 0.0648 m/day. With n = 0.25, v = 0.2592 m/day; 100 m travel $\approx$ 386 days.

Specific yield + specific retention = porosity in the course table. Yield drains; retention remains held.

Flow-direction check: use head, not ground elevation alone. The negative sign in the full Darcy form indicates flow toward decreasing head; calculation questions often request the positive magnitude.

SURFACE-WATER DISCHARGE [M10]

Subsection area: A_i = w_id_i. **Mean velocity:** $\bar{v}_i = (v_{0.2D} + v_{0.8D})/2$.

Discharge: Q = A_i $\bar{v}_i$; Q_{total} = ΣQ_i . m³/s = m³/s. Use ~20-30 slices, each ~3%-5% of total width. One velocity at 0.6D is an acceptable but less accurate substitute for 0.2D + 0.8D.

Worked course sum: seven 1 m-wide subsections produce Q values 1.500, 2.625, 4.000, 4.000, 4.000, 2.625, 1.500 m³/s; 2Q = **20.25 m³/s**.

Rate-volume check: 26.41 m³/s $\times$ 3600 $\times$ 24 = 31 = 70,736,544 m³ in December if flow is constant.

Stage: H_{stage} = H - H₀. A rating curve converts stage to discharge and is valid near the section where calibrated.

Rating checks: water surface 155 m and datum/bottom 140 m $\rightarrow$ stage 15 m. Course curve example: stage 30 m estimates Q $\approx$ 70 m³/s.

Return period: annual probability P = 1/T; percentage = 100/T. Example: T = 50 years $\rightarrow$ 2% chance every year, not once every 50 years.

Other return checks: T = 100 years $\rightarrow$ 1% annual chance; T = 20 years $\rightarrow$ 5%; T = 10 years $\rightarrow$ 10%. Independent annual risk repeats each year.

Hydrograph: discharge versus time; may overlay precipitation. Stormflow/quickflow = event response above baseflow. Urbanization makes a higher peak, steeper rising limb, and shorter lag.

NUMERICAL + GRAPH TRAPS

- Keep rate and volume distinct. A hydrograph ordinate is usually discharge, not stored volume.
- Preserve b's sign in groundwater ET.
- Use actual vertical water-column height for pressure.
- Darcy q and pore velocity v differ when n < 1.
- Read saturation vapour pressure at the stated temperature.
- Read a rating curve from stage to curve to discharge.
- Return period is probability, never a schedule.

VARIABLE KEY: GROUNDWATER + RIVERS

Elevation/head: z_{wt} = water-table elevation; z_{ground} = ground elevation; d_{water} = water-table depth below ground; z_b = well-bottom elevation; d_{well} = well depth; h = total hydraulic head; z = elevation head; h_p = pressure head.

Groundwater flow: p = fluid pressure; ρ = water density; g = gravitational acceleration; i = hydraulic gradient; Δh = head difference; L = flow-path length; K = hydraulic conductivity; q = Darcy specific discharge; Q = total discharge; A = flow area; v = pore-water velocity; n = porosity; t = travel time.

Channel slices: A_i = subsection area; w_i = width; d_i = representative depth; D = total local water depth; v_{0.2D}, v_{0.8D} = velocities at 20%/80% depth; $\bar{v}_i$ = subsection mean velocity; Q = subsection discharge; Q_{total} = summed channel discharge.

Stage/flood: H_{stage} = stage above datum; H = water-surface elevation; H₀ = local datum elevation; T = return period in years; P = annual exceedance probability. Context distinguishes P = precipitation from P = probability.

RECURRING COURSE LOGIC [M1-M12]

Scientific method: observation/question → testable hypothesis → prediction → controlled evidence → analysis → conclusion/revision. Conclusions remain provisional. **Boundary + scale:** the same conservation logic applies to a leaf, lysimeter, lake, watershed, aquifer, glacier, or Earth. A term can change classification when the boundary changes. **Storage vs. flux:** lake level, soil moisture, groundwater, snowpack, and glacier ice are storages; P, ET, runoff, recharge, and discharge are transfers. **Energy + gradients drive flow:** solar energy drives phase changes and ET; pressure/elevation gradients drive groundwater; channel gradient drives rivers; gravity drives glacier flow. **Measurement theme:** every instrument samples a place and time. Accuracy depends on siting, calibration, density, representativeness, and matching method to scale.

HYDROLOGY + WATERSHEDS [M2]

Physical hydrology studies water occurrence, movement, and properties in solid, liquid, and vapour forms. Major categories: atmospheric water, surface water, groundwater, cryosphere. Oceans hold ~97% of all water; freshwater is under 3%; most freshwater is frozen or underground. Glaciers/ice caps are the largest freshwater reserve.

Watershed/catchment/drainage basin: topographically bounded area draining to a defined outlet. **Divide:** boundary. **Tributary:** smaller stream entering larger. Subwatersheds nest.

Infiltration: entry into soil. **Percolation:** downward subsurface movement toward saturated zone. **Recharge:** addition to groundwater. **Runoff:** water routed over/near surface to channels.

Hydrographs show timing and magnitude of response. Baseflow persists between storms; stormflow is event response. Moving water transports sediment and contaminants along pathways away from sources.

PRECIPITATION PROCESSES [M3]

Course broad precipitation forms include rain, drizzle, snow, hail, sleet, fog, dew, frost. Liquid enters active flow quickly; solid water is stored until melt. **Convective:** surface heating, intense/local. **Orographic:** terrain uplift, wet windward + dry rain shadow. **Convergence:** colliding air masses rise. **Frontal:** warm/cold air boundaries.

Cold front = steeper, shorter, more intense rain. Warm front = gentler, longer, less intense rain. Frontal lifting dominates southern Ontario mid-latitudes. Equator: warm rising air, condensation, heavy rain. Near 30° N/S: descending warming dry air, arid belts. Earth's rotation and equator-to-pole temperature contrast drive circulation.

Rising air expands and cools adiabatically without losing heat. Condensation needs saturation plus nuclei such as dust/salt. Arid region course threshold: ≤25 cm annual P; deserts may be hot or polar.

PRECIPITATION MEASUREMENT [M4]

Non-recording gauge: manual total, no intensity history. **Recording:** weighing, float, tipping-bucket. Tipping bucket gives timing/intensity but can miss drops insufficient to tip. Errors: wind under-catch, evaporation, splashing, poor placement, observer/mechanical failure, station moves/equipment changes. Site in flat open low-wind area, at least 4× obstacle height away. Snow gauges are elevated, shielded, and melt snow to water equivalent. Arithmetic mean is simplest; Thiessen area-weights gauges; isohyetal bands best represent spatial/topographic variation.

EVAPORATION + ET CONCEPTS [M5-M6]

Evaporation needs energy and positive VPD. Increasing albedo reflects more radiation and tends to decrease evaporation. High humidity lowers VPD; wind renews dry air; high temperature raises e. Evapotranspiration includes free-water, interception, and soil evaporation, sublimation where applicable, and plant transpiration. Condensation of dew/frost is opposite-direction phase change, not ET. Seasonality in Canada: evaporation often high in spring/fall; transpiration peaks in summer and approaches zero in winter; total ET peaks in summer. Pan/lysimeter/chamber/energy balance/water balance/model/GW fluctuation each answer different scales. More decimal precision does not remove systematic or spatial error.

SOILS [M8]

Soil = mineral + organic matter + water + air. Regolith includes soil and rock over bedrock. Formation controls: parent material, time, climate, organisms, topography. Profile top to bottom: O, A, E, B, C, R. O+A = topsoil; B begins subsoil; R = bedrock. Above water table is vadose/unsaturated zone; course figure groups capillary fringe with saturated materials. **Weathering** breaks material down; **erosion** removes/transportes it. Mechanical weathering retains composition; chemical weathering creates different minerals. Higher water/wind/ice velocity generally increases erosion. Degradation: erosion, desertification, contamination, topsoil deterioration, fertilizer overuse, intensified agriculture. Controls: windbreaks, terraces, reduced/contoured slopes. Course soil characterization uses colour, texture, structure, pH. Consistency states and Atterberg limits track strength as water content changes.

GROUNDWATER SYSTEMS [M9]

Water table = top of saturated zone. Aquifer stores/transmits water; aquitard/confining layer restricts flow. Types: unconfined, confined, semi-confined/leaky, karstic. Confined water can rise to the potentiometric surface; if above ground, a flowing artesian well results. Piezometers map head and temporal change. Sand/gravel glacial deposits form major southern Ontario aquifers. Moraines store recharge and feed springs/streams. Karst and fractures can create rapid contaminant pathways. Stream interactions: gaining stream receives GW; losing stream supplies GW; perched losing stream is separated from regional water table; flow-through enters one bank and exits another. Sources: landfills, septic systems, farms/manure, road salt, spills, pesticides, pharmaceuticals, saline intrusion. Walkerton: manure runoff entered open Well 5; fractured karst connected aquifers. **Accuracy theme:** groundwater cannot be seen directly; infer it from wells, head, geology, K, tracers, and models. One well is local evidence, not a complete aquifer map.

HIGH-FREQUENCY DISTINCTIONS

Potential vs actual ET; evaporation vs transpiration vs sublimation; infiltration vs percolation; porosity vs permeability; yield vs retention; q vs v; unconfined vs confined; weathering vs erosion; storage vs rate.

RUNOFF, RIVERS, FLOODS [M10]

Runoff components: direct precipitation, shallow interflow, groundwater discharge, snowmelt. Controls: climate, catchment size, P type/intensity/distribution, hydraulic structures, vegetation, urbanization. Urban impervious cover reduces infiltration, speeds routing, raises peak discharge and flood risk, and shortens lag. Vegetation/interception/storage generally delay and reduce runoff. Natural streamflow is mainly turbulent. Velocity is lowest near rough bed/banks and highest just below surface along main flow; in meanders, fastest water shifts outside. Outside meander bank: faster flow + erosion. Inside: slower flow + deposition. Cutoff creates oxbow lake on floodplain. Gradient is usually steepest near headwaters. **Sediment:** bed load by traction/saltation; fine particles suspended; ions dissolved. **Competence** = largest particle carried. **Capacity** = total load. Fine cohesive clay may require high erosional velocity. Stilling well + float/shaft encoder gives stable stage. Flood controls: structural works and non-structural land-use, warning, and floodplain planning.

GLACIERS + GLACIATION [M11]

Glacier = persistent compacted/recrystallized snow flowing under its weight. Positive budget: accumulation > ablation; negative: ablation > accumulation. Equilibrium line separates zones. Alpine ice is topographically confined; continental ice sheets cover vast land and ignore small obstacles. Glacier can flow downslope while the toe retreats upslope. Movement: internal plastic flow is primary; basal slip uses meltwater lubrication. Velocity depends on slope, ice volume, temperature; normal 0-4 m/day, surges 10-100×. Milankovitch: eccentricity = orbit shape; obliquity = axial tilt (~23.5° now); precession = axis direction. Together alter insolation and help explain glacial cycles. Last glacial maximum ~20,000 years ago.

Erosional alpine: U-shaped valley, hanging valley, cirque, arete, horn. **Depositional:** till/moraines. Lateral = side; medial = joined centre; terminal/end = toe; recessional = retreat pauses. **Continental/meltwater:** drumlin = streamlined hill/ice direction; esker = sinuous sand-gravel ridge; kame = conical sediment hill; kettle = water-filled depression; outwash = meltwater sediment.

Isostatic depression = loading sinks crust; rebound = slow uplift after melt (~1 cm/year Hudson Bay). Glacial valleys are U-shaped, not V-shaped.

CLIMATE CHANGE + HYDROLOGY [M12]

Climate requires ≥30 years; a single event/year is weather. Greenhouse atmosphere transmits incoming shortwave more readily and absorbs outgoing longwave, warming air, land, and ocean. Important course gas: CO₂; also CH₄, nitrous oxide, CFCs. Natural drivers: solar output, orbit, plate tectonics, volcanoes. They do not explain rapid current warming; human annual CO₂ ~60× volcanic. **Quiz 12 anchors:** doubled current CO₂ → approximately +3°C global mean warming (±1°C). The saved quiz pairs its stated 1867 fossil-fuel/CO₂ warming work with Svante Arrhenius. Evidence: instruments + tree rings, corals, sediments, ice cores; Keeling Curve since 1958; ice cores span 800,000 years. CO₂ ~280 ppm preindustrial and just under 420 ppm in July 2022.

Documented hydrologic changes: warmer surface water, less ice cover, earlier snowmelt, increased high-latitude runoff, heavier P events, shorter lake/river ice seasons, sea-level rise. Sea level almost 4 mm/year vs 1-2 mm/year early 20th century and accelerating. Constant-rate baseline: Δh = rt. IPCC = Intergovernmental Panel on Climate Change. Mitigation reduces causes/emissions; adaptation manages impacts. Climate models with natural + human forcing match observed warming better than natural-only runs.

LAST-MINUTE TRUE/FALSE TRAPS

Steady state can have nonzero flows: **T**. Oxbow lakes occur on floodplains: **T**. Max channel velocity is right at surface: **F** (just below). Capacity = largest grain: **F** (competence). Isostatic rebound is sinking: **F**. Alpine glaciers cut V valleys: **F**. Glaciers are largest freshwater reserve: **T**. Hydrograph may show P + Q: **T**. Climate-scale minimum is 30 years: **T**.